

Algebraic expressions

A letter standing for a number — and a sentence in words turned into one line of algebra.

LESSON 8–10

3 × 40 MIN

MATEMATIKA 6, §1

S7 2.1

LESSON CLOCK

20'

30'

30'

30'

10

A letter for a number	20 min
Words into algebra	30 min
Algebra into words	30 min
The shorthands	30 min
Homework	10 min

LEARNING OBJECTIVES

- Explain what a letter stands for in an expression.
- Write a worded rule as an algebraic expression.
- Read an expression back into words.
- Use the standard shorthands of algebraic notation.

TEXTBOOK

National	pp. 21–28
Cambridge	Stage 7, pp. 20–24
Workbook	Exercise 2.1

01

Explanation

A letter for a number

An **algebraic expression** is an arithmetic expression in which some numbers are replaced by letters. The letter stands for a number that may change — which is why it is called a **variable**.

EXPRESSION	WHAT THE LETTER STANDS FOR
$3n$	the number of legs on n tripods
$x + 5$	a number five more than x
$2(a + b)$	the perimeter of a rectangle a by b
$100 - p$	the change from 100 after spending p

WHY LETTERS ARE WORTH THE TROUBLE

One line, $P = 2(a + b)$, replaces a sentence that would have to be rewritten for every rectangle in the world. Algebra is arithmetic said once instead of endlessly.

Words into algebra

IN WORDS	IN ALGEBRA
a number increased by 7	$x + 7$
a number decreased by 7	$x - 7$
7 less than a number	$x - 7$
a number subtracted from 7	$7 - x$
three times a number	$3x$
a third of a number	$\frac{x}{3}$
three times a number, then add 2	$3x + 2$
add 2 to a number, then treble it	$3(x + 2)$

THE ORDER OF THE WORDS DECIDES THE BRACKETS

$3x + 2$ and $3(x + 2)$ are different: for $x = 4$ they are 14 and 18. “Then” tells you which operation happens second, and a bracket is how you write that.

Algebra into words

Reading in the other direction is the better test of understanding — and there is usually more than one right reading.

EXPRESSION	ONE READING	A CONTEXT
$5n$	five times a number	the cost of n pens at 5 each
$n + 3$	three more than a number	an age three years from now
$2n + 1$	double a number and add one	an odd number
$\frac{s}{t}$	one number divided by another	speed, from distance and time

EVERY FORMULA IS AN EXPRESSION WITH A NAME

$P = 2(a + b), S = ab, v = \frac{s}{t}$ — the left side names what the expression on the right computes.

The shorthands

WRITTEN AS	NOT AS	RULE
$3x$	$3 \cdot x$	the multiplication sign is dropped
ab	$a \cdot b$	the same between two letters
x	$1x$	a coefficient of 1 is not written
$-x$	$-1x$	the same for -1
$\frac{x}{3}$	$x \div 3$	division is written as a fraction
x^2	$x \cdot x$	a power counts equal factors

3X MEANS MULTIPLY, NEVER “THIRTY-SOMETHING”

In arithmetic, digits written together mean place value; in algebra, symbols written together mean a product. It is the one place where the two notations disagree.

02 Worked examples

EXAMPLE 1 Write “three times a number, then add 2” and “add 2 to a number, then treble it”.

- ① First: multiply, then add.

$$3x + 2$$

- ② Second: add, then multiply — the addition needs a bracket.

$$3(x + 2)$$

- ③ They differ: for $x = 4$ they give 14 and 18.

ANSWER

$$3x + 2 \text{ and } 3(x + 2)$$

EXAMPLE 2 A pen costs p sum and a pencil q . Write the cost of 3 pens and 4 pencils.

- ① 3 pens cost $3p$.

- ② 4 pencils cost $4q$.

- ③ Together: $3p + 4q$.

ANSWER

$$3p + 4q$$

EXAMPLE 3 Write the perimeter and area of a rectangle with sides a and b .

- 1 Perimeter: two lots of $a + b$.

$$P = 2(a + b)$$

- 2 Area: one side times the other.

$$S = ab$$

ANSWER

$$2(a + b) \text{ and } ab$$

03 Interactive model

Read an expression aloud and ask for a story that fits it; the class invents contexts far faster than they translate them, and the translation improves as a result.

Words into algebra

“A number increased by 7”:

$7x$

$x + 7$

$x - 7$

$7 - x$

“7 less than a number”:

$7 - x$

$x - 7$

$x + 7$

$7x$

“A number subtracted from 7”:

$7 - x$

$x - 7$

$x + 7$

$7x$

“Three times a number, then add 2”:

$$3x + 2$$

$$3(x + 2)$$

$$2x + 3$$

$$x + 6$$

“Add 2 to a number, then treble it”:

$$3x + 2$$

$$3(x + 2)$$

$$2(x + 3)$$

$$3x + 6 \text{ only}$$

“A third of a number”:

$$3x$$

$$\frac{x}{3}$$

$$x - 3$$

$$x + 3$$

$1x$ is written:

$$1x$$

$$x$$

$$x^1$$

$$x + 1$$

The perimeter of an a by b rectangle:

$$ab$$

$$a + b$$

$$2(a + b)$$

$$2ab$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Algebraic expression	Harfli ifoda	Буквенное выражение
Variable	O'zgaruvchi	Переменная
Constant	O'zgarmas	Постоянная
Term	Had	Член
Coefficient	Koeffitsiyent	Коэффициент
Sum	Yig'indi	Сумма
Product	Ko'paytma	Произведение
Formula	Formula	Формула

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A letter in an expression stands for:

a word

a number

an operation

nothing

$3x$ means:

$3 + x$

$3 \cdot x$

thirty-something

x^3

“7 less than x ” is:

$7 - x$

$x - 7$

$7x$

$x + 7$

$3(x + 2)$ at $x = 4$ is:

14

18

20

12

A third of x is written:

$3x$

$\frac{x}{3}$

$\frac{3}{x}$

$x - 3$

The area of an a by b rectangle:

$2(a + b)$

ab

$a + b$

$2ab$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. “A number increased by 7”

ANSWER $x + 7$

2. “A number decreased by 7”

ANSWER $x - 7$

3. “Three times a number”

ANSWER $3x$

4. “A third of a number”

ANSWER $\frac{x}{3}$

5. “A number subtracted from 7”

ANSWER $7 - x$

6. The cost of n pens at 5 sum

ANSWER $5n$

7. The perimeter of an a by b rectangle

ANSWER $2(a + b)$

Medium · the standard question

7 PROBLEMS

1. “Three times a number, then add 2”

ANSWER $3x + 2$

2. “Add 2 to a number, then treble it”

ANSWER $3(x + 2)$

3. 3 pens at p and 4 pencils at q

ANSWER $3p + 4q$

4. The change from 100 after spending p

ANSWER $100 - p$

5. An age n now, in 5 years

ANSWER $n + 5$

6. An odd number, written with n

ANSWER $2n + 1$

7. Speed from distance s and time t

ANSWER $\frac{s}{t}$

Hard · stretch and reason

7 PROBLEMS

1. The perimeter of a square of side a

ANSWER $4a$

2. The total of three consecutive numbers starting at n

ANSWER $3n + 3$

3. The cost of n books at p each with 2000 postage

ANSWER $np + 2000$

4. Half of one number added to twice another

ANSWER $\frac{a}{2} + 2b$

5. The area of a rectangle whose length is 3 more than its width w

ANSWER $w(w + 3)$

6. Read $2(n - 1)$ in words

ANSWER One less than a number, then doubled

7. Why do $3x + 2$ and $3(x + 2)$ differ?

ANSWER The bracket changes which operation is done first

07 Homework

Homework — 5 tasks

Say the expression aloud before writing it; the word “then” marks the bracket.

- Write in algebra: a number increased by 12; 12 less than a number; a quarter of a number.
- Write “double a number, then subtract 5” and “subtract 5 from a number, then double it”.
- A notebook costs a and a pen b . Write the cost of 5 notebooks and 2 pens.
- Write the perimeter and area of a square of side a .

5. Read $4(n + 3)$ in words and give its value at $n = 2$.